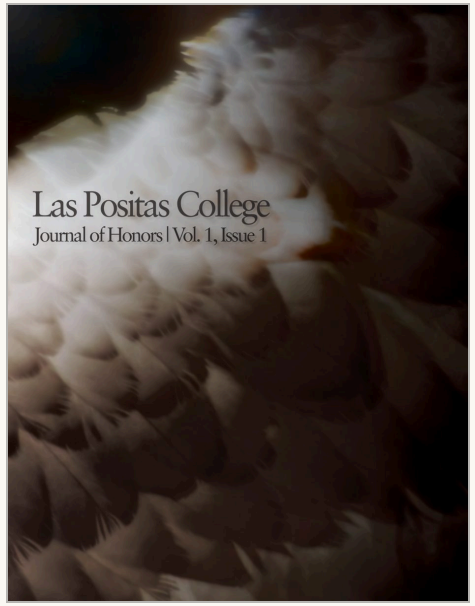
Theme Research

Two editorial directions for the next issue: In Between and Fragments. Each has a cover, a section divider and accents for research pages.



Nikhilesh Suravarjjala
November 2026

What the first issue already establishes



Cover, p. 1

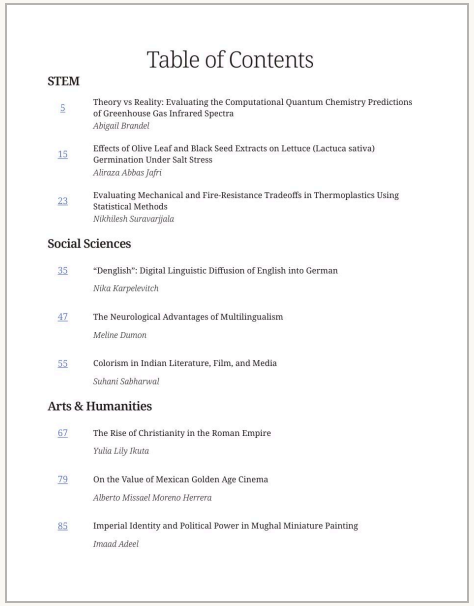


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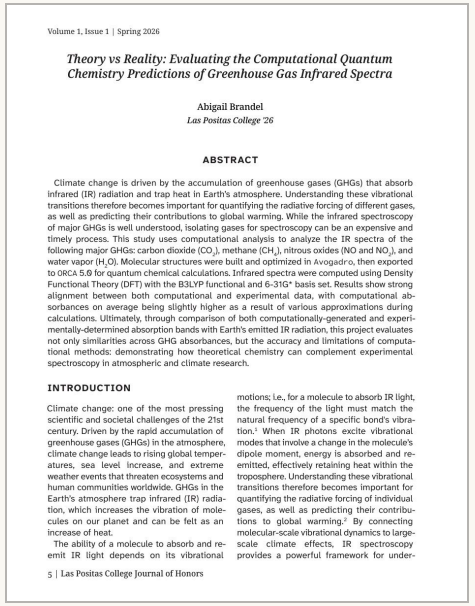
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Volume 1, Issue 1 | Spring 2026

Theory vs Reality: Evaluating the Computational Quantum Chemistry Predictions of Greenhouse Gas Infrared Spectra

Abigail Brandel
Las Positas College '26

ABSTRACT

Climate change is driven by the accumulation of greenhouse gases (GHGs) that absorb infrared (IR) radiation and trap heat in Earth's atmosphere. Understanding these vibrational transitions therefore becomes important for quantifying the radiative forcing of different gases, as well as predicting their contributions to global warming. While the infrared spectroscopy of major GHGs is well understood, isolating gases for spectroscopy can be an expensive and timely process. This study uses computational analysis to analyze the IR spectra of the following major GHGs: carbon dioxide (CO_2), methane (CH_4), nitrous oxides (NO and NO_2), and water vapor (H_2O). Molecular structures were built and optimized in Avogadro, then exported to ORCA 5.0 for quantum chemical calculations. Infrared spectra were computed using Density Functional Theory (DFT) with the B3LYP functional and 6-31G* basis set. Results show strong alignment between both computational and experimental data, with computational absorbances on average being slightly higher as a result of various approximations during calculations. Ultimately, through comparison of both computationally-generated and experimentally-determined absorption bands with Earth's emitted IR radiation, this project evaluates not only similarities across GHG absorbances, but the accuracy and limitations of computational methods: demonstrating how theoretical chemistry can complement experimental spectroscopy in atmospheric and climate research.

INTRODUCTION

Climate change: one of the most pressing scientific and societal challenges of the 21st century. Driven by the rapid accumulation of greenhouse gases (GHGs) in the atmosphere, climate change leads to rising global temperatures, sea level increase, and extreme weather events that threaten ecosystems and human communities worldwide. GHGs in the Earth's atmosphere trap infrared (IR) radiation, which increases the vibration of molecules on our planet and can be felt as an increase of heat.

The ability of a molecule to absorb and re-emit IR light depends on its vibrational motions; i.e., for a molecule to absorb IR light, the frequency of the light must match the natural frequency of a specific bond's vibration.[1] When IR photons excite vibrational modes that involve a change in the molecule's dipole moment, energy is absorbed and re-emitted, effectively retaining heat within the troposphere. Understanding these vibrational transitions therefore becomes important for quantifying the radiative forcing of individual gases, as well as predicting their contributions to global warming.[2] By connecting molecular-scale vibrational dynamics to large-scale climate effects, IR spectroscopy provides a powerful framework for under-

5 | Las Positas College Journal of Honors

Research page, p. 5

Verified in the PDF

112 pages, US Letter (612 × 792 pt), exported from Affinity.

The cover is one full-page photograph; the masthead reads "Las Positas College / Journal of Honors | Vol. 1, Issue 1" in a light serif.

Contents groups work as STEM, Social Sciences, Arts & Humanities, Op-eds. There are no divider pages: each group starts on an article.

Research pages: two columns, about 53 pt side margins, Atkinson Hyperlegible Next at 12 pt, Noto Serif italic titles, folio "5 | Las Positas College Journal of Honors".

Carried forward

Masthead wording and light serif, folio and running-header format, Letter size. New: section dividers, a colour system, a display face.

In Between

The spaces separating ideas are also where ideas meet.

Why it fits

Every article in an issue sits on a border: chemistry and climate, language and social media, history and art. The theme treats the border as the subject.

Colour theory

Cool analogous violets carry the structure, with one warm coral counterpoint. Lavender sits between indigo and iris as a bridge tone. Colour describes transition: overlaps, gradual steps, translucent planes.



Midnight Indigo	Electric Iris	Soft Lavender	Living Coral	Warm Ivory
#25235C · ~35%	#4948ED · ~25%	#A29CF4 · ~15%	#FF755B · ~10%	#F3EDE3 · ~15%
Structural foundation	Primary expressive colour	Transition and visual bridge	Warm focal counterpoint	Neutral editorial space



Two planes cross. Where they overlap, the work is shared.

The crossing

Above the coral nexus the planes overlap in lavender. Below it they part and leave an ivory gap.

Masked title

IN and BETWEEN switch ink wherever the plane under a letter changes, so the title is cut by its own subject.

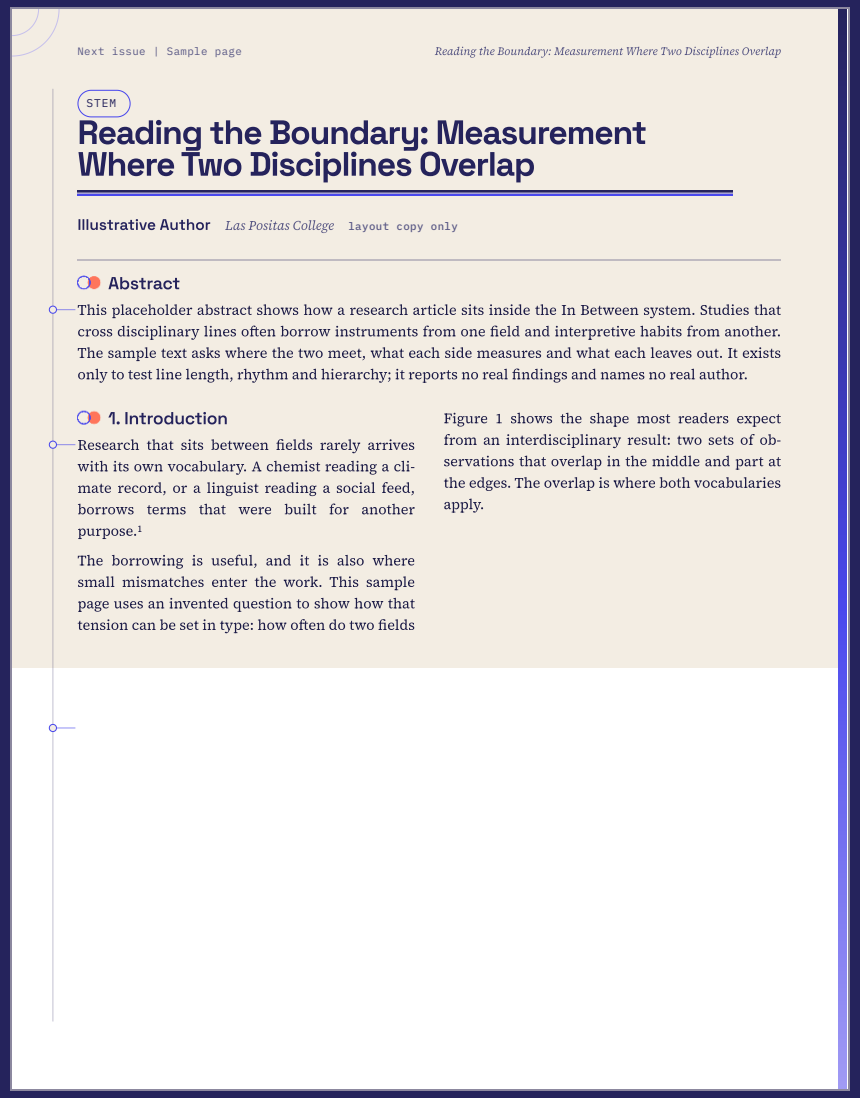
Construction

Concentric arcs and a crosshair run through the nexus. Every hairline is clipped and re-inked per region.

Masthead

“Las Positas College / Journal of Honors” stays in a light serif on an ivory band, as on Spring 2026.





Divider

S, T, E and M each sit in a circle. Every intersection takes a new colour, so the fields visibly share letters.

Heading marker

Two overlapping circles, one outlined and one coral.

Rules and capsules

Dual-tone title rule, outlined figure capsule, margin connector line.

Gradient edge

An 8 px indigo to lavender strip. The only gradient on the page.

Fragments

Knowledge is assembled from pieces that rarely arrive together.

Why it fits

Few studies are complete alone. Each article adds a piece, and the issue is where the pieces are put together.

Colour theory

Saturated primaries separated by hard edges. Citrine for discovery, Vermilion for interruption, Cobalt for a cool counterweight. Ink and Paper hold the page. No gradients.



Editorial Ink	Citrine Yellow	Vermilion	Cobalt Blue	Warm Paper
#191919 · ~30%	#F2D84A · ~17%	#E44F3F · ~10%	#3873D9 · ~13%	#F5F0E5 · ~30%
Structural typography	Discovery and attention	Energy and interruption	Cool counterweight	Neutral editorial foundation



One word, cut into strips and put back one column off.

12 × 16 grid

Every block and every displacement lands on a 60 px module, or half of one.

Colour blocks flip ink

Letters turn paper, ink or white as they cross cobalt, citrine, vermilion and ink fields.

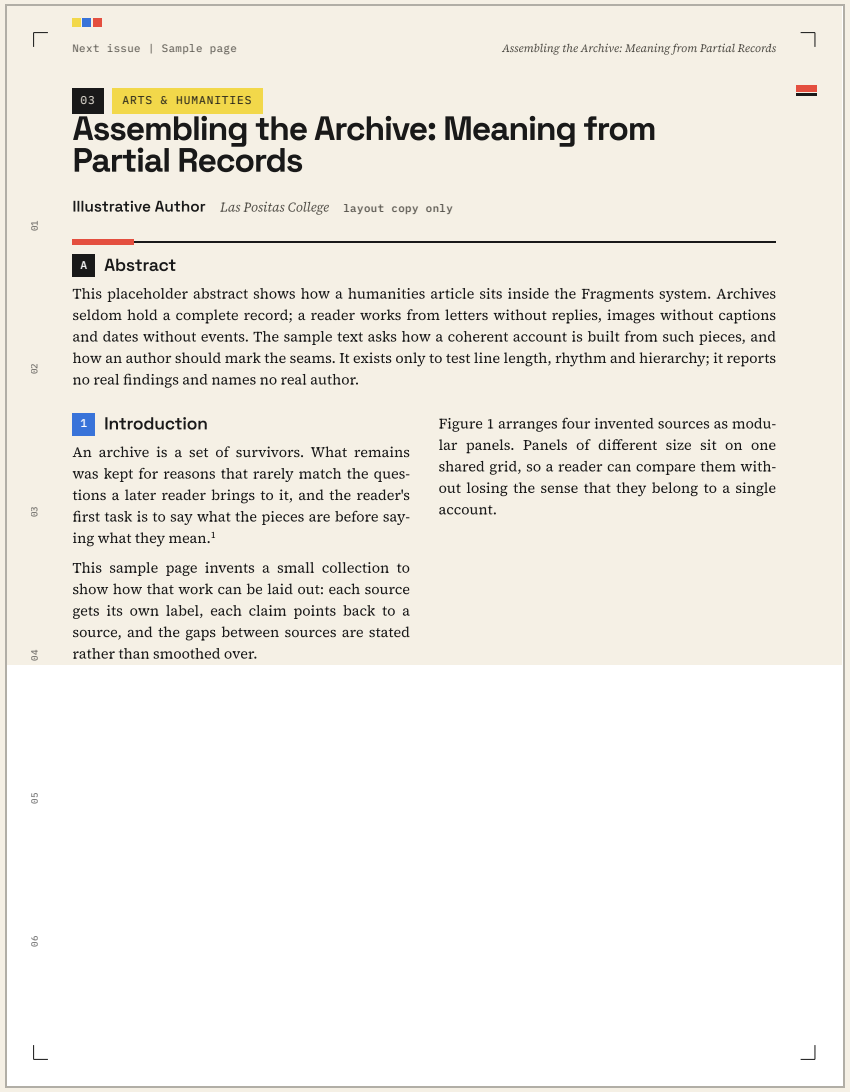
Numbered fragments

01 to 04 label the blocks; the panel at lower right shows the parts assembled.

Masthead

Same light serif lockup, set paper-on-ink at the base.





Divider

A, R, T and S step down the grid in four blocks, each overlapping the last by one cell.

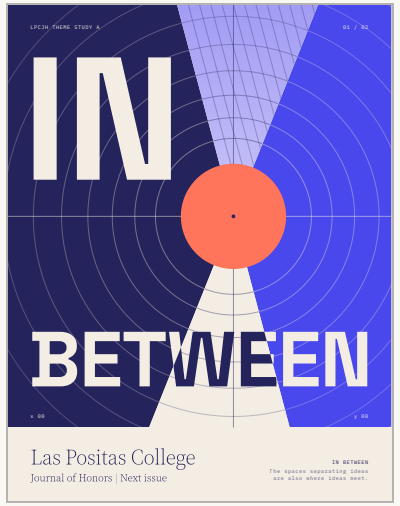
Tags and labels

Numbered section tag, square heading markers, “FIG” label with a colour chip.

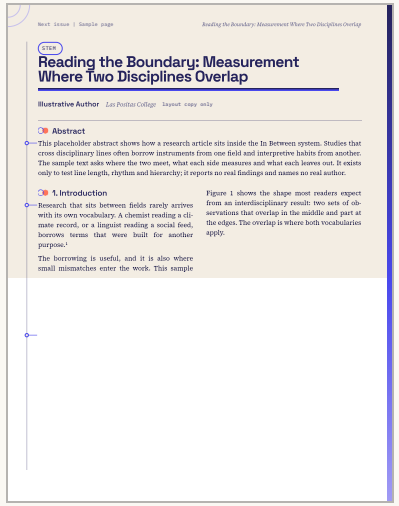
Margin details

Crop marks, a colour strip and a folio on an offset block. All outside the text.

Same journal, two systems

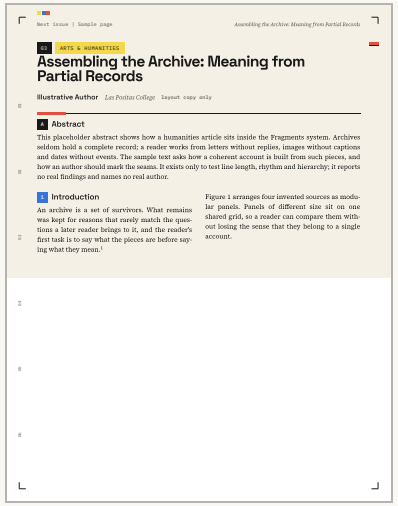












	In Between	Fragments
Narrative	Boundaries are where ideas meet.	Knowledge is assembled from pieces.
Colour	Analogous violets, one warm counterpoint, gradual change.	Primaries split by hard edges, no gradients.
Geometry	Crossing planes, circles, arcs.	Blocks, strips, split circles on a 60 px grid.

Flexibility, accessibility, production

	In Between	Fragments
Editorial flexibility	Dividers for the other sections reuse the circle-and-intersection method with different letter sets. Tone stays calm, which suits longer reading.	Any word breaks into strips or blocks. Higher energy; a long title needs more care to stay legible.
Visual accessibility	Body text 13.8:1 on ivory. Ivory on lavender is 2.1:1, so that pairing is decorative only; indigo on lavender is 5.8:1.	Body 15.5:1 on paper. Cobalt takes white (4.5:1) for small labels; paper on cobalt (4.0:1) is for display type only.
Production	All vector. One gradient and a few transparencies; both export cleanly to PDF.	All vector, flat fills only. The grid is easy to rebuild in InDesign or Affinity.
Type	Shared: Space Grotesk, Source Serif 4, IBM Plex Mono (all open licence). The Spring 2026 issue sets body text in Atkinson Hyperlegible Next; swapping it back is a one-token change if readability policy requires it.	